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Department Of Information Technology

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Class: BE-ITA/B, Semester: VIII

Subject: BlockChain Lab

Experiment – 1

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1. **Aim:** Case Study on Blockchain
2. **Objectives:** Students should be able to understand application areas of Blockchain Technology in real world.
3. **Prerequisite:** Distributed systems and Security basics
4. **Requirements:** Online/Library resources
5. **Pre-Experiment Exercise:**

• **Theory:** Write Case study on Following points

- Define Blockchain
- Types of Blockchain
- Features of Blockchain
- Security in Blockchain

Laboratory Exercise

7. **Procedure**

- A. □ Refer to any authenticated web reference to complete the following tables

S.no	Application Domain	Real Life Application
1.	Banking	
2	Healthcare	
3	Taxation	
4	Insurance	
5	Gaming	
6	Mobile Payment	
7	IOT	
8	Governance	
9.	Border security	
10	Music	Example: Arbit: It is a blockchain based project led by former Guns N Roses drummer Matt Sorum seeking a fairer way to reward musicians for their creative efforts.

- Find and Study opensource tools used for blockchain applications.

S.no	Assignment	Open-Source Tool
1.	To implement basic hash chain	
2	Basic Ehereum Transactions	
3	Implement Smart Contracts	
4	Languages used for Blockchain platform	

· **Post-Experiments Exercise:**

· **Extended Theory:**

- Advantages of Blockchain
- Limitations and Challenges of Blockchain

9. **Conclusion:**

Block chain technology has capability to transform every traditional industry to a new one either it is financial or non-financial.

10. **References:**

- [1] Any online /offline reference

Theory

Case Study:

Define Blockchain:

Blockchain is a shared, immutable digital ledger, enabling the recording of transactions and the tracking of assets within a business network and providing a single source of truth.

Blockchain operates as a decentralized distributed database, with data stored across multiple computers, making it resistant to tampering. Transactions are validated through a consensus mechanism, ensuring agreement across the network.

In blockchain technology, each transaction is grouped into blocks, which are then linked together, forming a secure and transparent chain. This structure guarantees data integrity and provides a tamper-proof record, making blockchain ideal for applications like cryptocurrencies and supply chain management.

Types of blockchain

There are several ways to build a blockchain network. They can be public, private, permissioned or consortium-built:

Public blockchain networks:

A public blockchain is open for anyone to join and participate in, such as the Bitcoin blockchain. While it offers decentralization, it also comes with drawbacks, including high computational power requirements, lack of transaction privacy and potentially weaker security. These considerations are crucial, especially for enterprise blockchain use cases.

Private blockchain networks:

A private blockchain network, similar to a public blockchain network, is a decentralized peer-to-peer network. However, one organization governs the network, controlling who is allowed to participate, run a consensus protocol and maintain the shared ledger. Depending on the use case, this infrastructure can significantly boost trust and confidence between participants. A private blockchain can be run behind a corporate firewall and even be hosted on premises.

Permissioned blockchain networks:

Businesses that set up a private blockchain set up a permissioned blockchain network. It is important to mention that public blockchain networks can also be permissioned. This places restrictions on who is allowed to participate in the network and in what record transactions. Participants need to obtain an invitation or permission to join.

Consortium blockchain networks:

A group of preselected organizations actively manages a consortium blockchain network and shares the responsibility of maintaining the blockchain. These organizations determine who can submit transactions and access data. This type of network is ideal when multiple parties must collaborate with shared responsibilities. In the energy sector, energy producers and consumers might share data about power usage and distribution.

Features of Blockchain

Blockchain technology contains several key features that enhance security, transparency and efficiency in transactions and data management:

Distributed ledger technology

All network participants have access to the distributed ledger and its immutable record of transactions. This shared ledger records transactions only once, eliminating the duplication of effort typical of traditional business networks.

Immutable records

No participant can change or tamper with a transaction after it's been recorded in the shared ledger. If a transaction record includes an error, a new transaction must be added to reverse the error, and both transactions are then visible.

Smart contracts

Smart contracts are self-executing agreements stored on the blockchain, where the terms are written in code and automatically executed when predefined conditions are met. They can be used for various purposes, such as transferring corporate bonds or triggering travel insurance payouts. By automating these processes, smart contracts speed up transactions, reduce the need for intermediaries and ensure transparency and security.

Public key cryptography

Public key cryptography is a method used to secure transactions and data on the blockchain by leveraging two cryptographic keys: a public key and a private key. The public key serves as an address for receiving cryptocurrency or

data, while the private key is a confidential key that grants control over the associated digital assets. The private key holder can authorize transactions, providing security and verifying ownership, while the public key allows others to send funds or data to the correct address.

Security in Blockchain

By virtue of its decentralized design and use of cryptography, blockchains are, in general, fairly secure. Once a block of data has been added to the chain and verified, it cannot be removed, and multiple blocks are always stored linearly (so it's easy to check the ledger for systematic problems). Equally, the records stored in most blockchains are encrypted, so sensitive data is often difficult to access even when hacked. However, blockchain networks are more complicated. Despite the ledger technology itself being theoretically "impenetrable", the network itself is not immune to cyberattacks and attempted fraud. Since their inception, there have been a number of different attacks on various blockchain networks.

51% Attacks

A particularly processing-intensive and power-intensive cyberattack, a 51% attack is instigated by a group of "miners" (originally part of the network) who leverage their combined resources to control enough of the network's mining power (more specifically, the network's mining hash rate). In doing so, they effectively gain control of the ledger itself. Once they have control of the ledger, the rogue miners can manipulate the transactions on it to enact financial fraud. Private networks are not usually susceptible to this type of attack.

Routing Attacks

A routing attack is very hard to spot because most of the attack happens behind the scenes. This attack involves intercepting data being transferred to your internet service provider, dividing the network, and targeting a certain chain between certain nodes. Essentially having created a kind of parallel chain, the attacker can then steal any currency or personal information from their targeted area.

Sybil Attacks

A Sybil attack involves a hacker using a node to create multiple fake identities or 'Sybil identities' to flood a network. This then allows them to carry out a 51% attack, see above, where they gain a 51% majority control of the network's computing power: thereby giving the hacker a disproportionately large influence or presence in the network. Common in distributed systems, it undermines trust and integrity, prompting defense through identity verification and cryptographic protocols.

Phishing Attacks

A classic online scamming technique that many users should be well-acquainted with, phishing attacks on a blockchain network are not necessarily an attack on the network in the same way that the last 3 examples are. A blockchain phishing attack targets the members of the blockchain with phishing emails to attain their credentials, with the intention of stealing currency from their accounts.

The role of cybersecurity, or more precisely, data security in blockchain networks is integral to today's publicly (and privately) used distributed ledger technologies. Below, we've combined a list of best practices and frameworks that both enterprises and network administrators can use to enhance their blockchain security moving forward.

Identity and Access Management: For both public and private networks, access controls and legitimizing communication between users and nodes are integral to the internal protection of sensitive data transfers and currency. This includes deciding whether block payloads are encrypted or not and how users' private and public keys are managed and revoked.

Governance and Risk Management: Consider what your disaster recovery plan for the blockchain participants is, if the worst should happen. Perform regular risk assessments to discover important vulnerabilities or other weaknesses in your chain. Equally, remember that blockchains are subject to all cybersecurity and privacy laws, regulations, and other nation-specific requirements throughout their lifecycle.

Use a proven VPN : Recommend that users and nodes activate a trusted and proven VPN when making transactions. A Virtual Private Network works by creating an encrypted private tunnel between a user's remote computer and any external servers, so your data or transactions will remain protected from any outside threats to the network.

Choose a Dedicated Antivirus Software: Overall, known threats to blockchain networks, particularly phishing scams, are

much easier to avoid if you have a dedicated antivirus software installed on your local machine. A good antivirus system will not only defend you from known threats but will also run regular scans to warn you of any new ones. Equally, antivirus software provides you with an extra layer of protection when you're validating transactions as a node. We recommend using our dedicated security software Kaspersky Premium, which offers regular updates, scans, and consistent help and support with all your digital activity.

Laboratory Exercise:

Procedure:

S.no	Application Domain	Real Life Application
1.	Banking	Kinexys, a private blockchain network for wholesale payments by JP Morgan & Chase.
2	Healthcare	The MediLedger blockchain is an industry-led, permissioned network for the life sciences, using blockchain to create trust and automate processes among pharma manufacturers, distributors, and health systems, primarily for DSCSA compliance (track-and-trace), contract management (chargebacks), and data integrity, enabling secure, shared data access for better supply chain visibility and reduced revenue leakage without compromising privacy using zero-knowledge proofs.
3	Taxation	Estonia and Singapore have leveraged blockchain for tax automation in their e-governance services.
4	Insurance	The RiskStream Collaborative is a non-profit consortium that uses a private blockchain network to help insurance companies securely share data and automate multi-party tasks like claims and death notifications. By creating a "single source of truth," it eliminates manual paperwork and reconciliation, significantly reducing administrative costs and fraud across the industry.
5	Gaming	Blockchain technology has numerous applications in gaming, primarily by introducing true ownership of digital assets, enabling play-to-earn (P2E) economies, and establishing decentralized, transparent systems. This fundamentally shifts control and value from developers to the players. E.g Roblox, Metaverse.
6	Mobile Payment	Ripple (The Company): A private fintech company that builds a global payment network called RippleNet . Its goal is to replace or upgrade the aging SWIFT system used by banks for international money transfers.
7	IOT	NetObjex operates through its Matrix Digital Asset-as-a-Service platform. It acts as an "operating system" for smart devices, following a specific workflow: IoT (Data Acquisition): Sensors collect real-time data from physical objects (e.g., a car, a shipping container, or a smart meter).
8	Governance	Nambikkai Inaiyam (Tamil for "Trust Link") is a first-of-its-kind state-wide blockchain backbone established by the Tamil Nadu e-Governance Agency (TNeGA). It serves as a secure, decentralized infrastructure for the entire state, allowing government departments to build "fraud-proof" digital services for citizens.

S.no	Application Domain	Real Life Application
9.	Border security	Stellar blockchain network for fast, secure, and transparent cross-border payments, rather than physical national border security.
10	Music	Example: Arbit: It is a blockchain based project led by former Guns N Roses drummer Matt Sorum seeking a fairer way to reward musicians for their creative efforts.

S.no	Assignment	Open-Source Tool
1.	To implement basic hash chain	HashLib
2	Basic Ethereum Transactions	Web3.js , Ganache
3	Implement Smart Contracts	Foundry,Truffle
4	Languages used for Blockchain platform	Python,Solidity, Rust